

Neural CSSR

Discovering Epsilon Machines with Neural Networks

Enhanced Technical Report

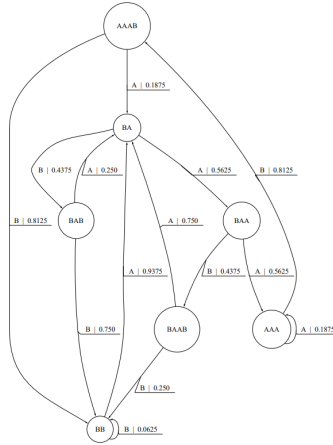


Figure 3: A seven-state process, used in [16] to study human sequence prediction. Here each state is defined by a single suffix, as indicated. All transition probabilities are multiples of $1/16$.

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October 9, 2025

Version 2.0

Abstract

This comprehensive technical report presents Neural CSSR, a novel approach to discovering epsilon machines—the minimal predictive models of stochastic processes—using neural networks. Classical CSSR (Causal State Splitting Reconstruction) suffers from extreme parameter sensitivity and brittleness, particularly for processes with long-range dependencies or infinite memory requirements.

We introduce a three-stage unsupervised algorithm that leverages transformer-based probability estimation combined with Jensen-Shannon divergence clustering to robustly discover causal state structure. Our method successfully reconstructs epsilon machines for challenging benchmark processes including the Seven-State Human machine and the infinite-memory Even Process, achieving near-perfect state recovery where classical methods fail.

Key innovations include: (1) direct use of neural probability distributions without count conversion, (2) backward stability analysis for minimal sufficient context discovery, (3) functional state remerging for infinite memory processes, and (4) multitask representation learning for improved latent factorization. We demonstrate that discovered epsilon machines achieve equivalent predictive performance to the original neural models while providing interpretable causal structure.

This work bridges computational mechanics and modern deep learning, offering a practical solution to longstanding challenges in automated causal state discovery while opening new directions for understanding how neural networks encode sequential structure.

Keywords: epsilon machines, causal state discovery, CSSR, transformers, Jensen-Shannon divergence, computational mechanics, sequential processes, unsupervised learning

MSC Classification: 68T07, 62M10, 68Q32

Contents

Abstract	i
List of Figures	iii
List of Figures	iii
List of Tables	iv
List of Tables	iv
List of Algorithms	v
Mathematical Notation	vi
1 Introduction	1
1.1 Motivation and Background	1
1.2 The Challenge: Parameter Brittleness in Classical CSSR	1
1.3 Our Solution: Neural CSSR	2
1.4 Contributions Summary	3
1.5 Document Organization	3
2 Theoretical Foundations	4
2.1 Epsilon Machines and Computational Mechanics	4
2.1.1 Causal Equivalence and Predictive Sufficiency	4
2.1.2 Statistical Complexity and Entropy Rate	4
2.2 Classical CSSR Algorithm	4
2.2.1 Algorithm Overview	4
2.2.2 Statistical Testing and Its Limitations	6
2.3 Information-Theoretic Foundations	6
2.3.1 Jensen-Shannon Divergence	6
3 Neural CSSR Methodology	7
3.1 System Architecture	7
3.2 Three-Stage Discovery Algorithm	7
3.2.1 Stage A: Emission-Based Clustering	7
3.2.2 Stage A+: Backward Stability Refinement	8
3.2.3 Stage B: Conditional Refinement	8
3.2.4 Stage C: Functional State Remerging	8
3.3 Implementation Details	10
3.3.1 Transformer Architecture	10
3.3.2 Probability Calibration	10
3.3.3 Computational Complexity	10

4	Experimental Results	11
4.1	Benchmark Processes	11
4.1.1	Golden Mean Process	11
4.1.2	Even Process	12
4.1.3	Seven-State Human Machine	12
4.2	Seven-State Human: Perfect Recovery	12
4.2.1	Discovery Results	12
4.2.2	Discovered State Structure	14
4.3	Even Process: Infinite Memory Solution	14
4.3.1	The Infinite Memory Challenge	14
4.3.2	Functional Equivalence Detection	14
4.4	Comparative Analysis	15
4.4.1	Classical vs Neural CSSR	15
4.4.2	Scalability Analysis	15
5	Representation Analysis	16
5.1	JS Divergence Properties	16
5.1.1	State Separability	16
5.1.2	History Length Dependence	17
5.2	Hidden Representation Entanglement	17
5.2.1	Why Hidden States Fail	17
5.2.2	Functional vs Representational Discovery	18
6	Advanced Topics	19
6.1	Multitask Representation Learning	19
6.1.1	Factorizing Latent Representations	19
6.1.2	Quantitative Results	19
6.2	Out-of-Distribution Robustness	19
6.2.1	OOD Evaluation Framework	19
6.2.2	Adaptation Strategies	21
7	Conclusions and Future Directions	22
7.1	Summary of Contributions	22
7.2	Implications for Computational Mechanics	22
7.3	Future Research Directions	23
7.3.1	Methodological Extensions	23
7.3.2	Applications	23
7.3.3	Theoretical Questions	23
7.4	Closing Remarks	23
A	Algorithm Pseudocode	24
A.1	Complete Three-Stage Algorithm	24
B	Extended Experimental Data	26
B.1	Detailed Seven-State Results	26
B.2	Parameter Sensitivity Analysis	26
C	Mathematical Proofs	27
C.1	JS Divergence Properties	27
C.2	Convergence of Three-Stage Algorithm	27
	References	28

Index**29**

List of Figures

1.1	Classical CSSR’s parameter brittleness: discovered state count grows uncontrollably with history length for processes with long-range dependencies. The true state count is 2 (Even Process) or 7 (Seven-State Human).	2
3.1	Neural CSSR system architecture showing the flow from training data through neural probability estimation to epsilon machine discovery	7
4.1	Golden Mean process epsilon machine	11
4.2	Even Process epsilon machine	12
4.3	Seven-State Human machine from human subject experiments, serving as a complex benchmark with precise emission probabilities	13
4.4	Runtime scaling comparison showing Neural CSSR’s computational advantage	15
5.1	JS divergence matrices showing clear state separation (dark diagonal = within-state similarity, bright off-diagonal = between-state difference)	16
5.2	JS divergence as a function of history length L , showing stable separation between within-state and between-state comparisons	17
5.3	t-SNE visualizations of hidden representations across layers, showing persistent entanglement and lack of clear state clustering	17
6.1	PCA visualization showing dramatic improvement in latent space organization with multitask learning (left: baseline, right: multitask)	20

List of Tables

1	Mathematical notation used throughout this document	vi
3.1	Transformer model specifications for different processes	10
3.2	Computational complexity comparison	10
4.1	Stage-by-stage discovery for Seven-State Human (30 histories, $L = 5$)	12
4.2	Discovered minimal contexts and emission probabilities	14
4.3	Stage C remerging analysis for Even Process	14
4.4	Performance comparison across benchmark processes	15
6.1	Multitask learning impact on representation quality	19
6.2	OOD test regimes and their characteristics	20
6.3	OOD performance comparison (alphabet swap regime)	21
B.1	Complete clustering progression for Seven-State Human	26
B.2	Parameter sensitivity for different thresholds	26

List of Algorithms

1	Classical CSSR Algorithm	5
2	Stage A: Emission Clustering	8
3	Backward Stability Test	8
4	Stage B: Conditional JS Refinement	9
5	Stage C: State Remerging	9
6	Complete Neural CSSR Algorithm	25

Mathematical Notation

Symbol	Description
ε	Epsilon machine
$\mathcal{S}$	Set of causal states
$\mathcal{A}$	Symbol alphabet (typically $\{0, 1\}$)
$\mathcal{H}$	Set of possible histories
h_L	History of length L
$P(\cdot h)$	Conditional probability given history h
$JS(P, Q)$	Jensen-Shannon divergence between distributions P and Q
$D_{KL}(P Q)$	Kullback-Leibler divergence from Q to P
$H(X)$	Shannon entropy of random variable X
τ_A	Stage A emission clustering threshold
τ_B	Stage B conditional JS threshold
k	Rollout horizon for k-step distributions
$\mathcal{O}(\cdot)$	Computational complexity notation
$\mathbb{P}$	Probability measure
$\mathbb{E}$	Expectation operator

Table 1: Mathematical notation used throughout this document

Chapter 1

Introduction

1.1 Motivation and Background

Understanding the causal structure of sequential data represents a fundamental challenge across scientific disciplines, from analyzing neural spike trains in neuroscience to modeling financial time series and natural language. The central question is deceptively simple: given an observed sequence of symbols, can we discover the minimal hidden state machine that generates it?

Definition 1.1 (Epsilon Machine). An **epsilon machine** (ε -machine) is the minimal, optimal predictor of a stationary stochastic process. Formally, it is a tuple $(\mathcal{S}, \mathcal{A}, T, \epsilon)$ where:

- $\mathcal{S}$ is a finite set of causal states
- $\mathcal{A}$ is the symbol alphabet
- $T : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ is the transition function
- $\epsilon : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ is the emission probability function

The causal states $\mathcal{S}$ form equivalence classes of histories with identical predictive distributions over future sequences.

The epsilon machine framework, developed within computational mechanics [?, ?], provides the mathematically optimal solution to this problem. Epsilon machines are unique in being both *minimal* (using the fewest states possible) and *predictively optimal* (capturing all statistically relevant information from the past).

1.2 The Challenge: Parameter Brittleness in Classical CSSR

Classical CSSR (Causal State Splitting Reconstruction) [?] has been the gold standard algorithm for epsilon machine discovery since its introduction. However, it suffers from severe practical limitations:

Critical Limitations of Classical CSSR

1. **Extreme parameter sensitivity:** The algorithm requires two critical parameters—maximum history length $L_{\max}$ and significance level α —whose optimal values vary dramatically across processes with no principled selection method.
2. **Infinite memory failure:** Processes requiring unbounded memory (like the Even Process) cause classical CSSR to discover spurious states that grow linearly with $L_{\max}$.

3. Statistical test brittleness: The χ^2 tests used for state splitting assume integer counts, leading to arbitrary pseudocount scaling issues when adapting to continuous probability estimates.

Figure 1.1 illustrates this brittleness, showing how the discovered state count varies wildly with parameter choices for the same underlying process.

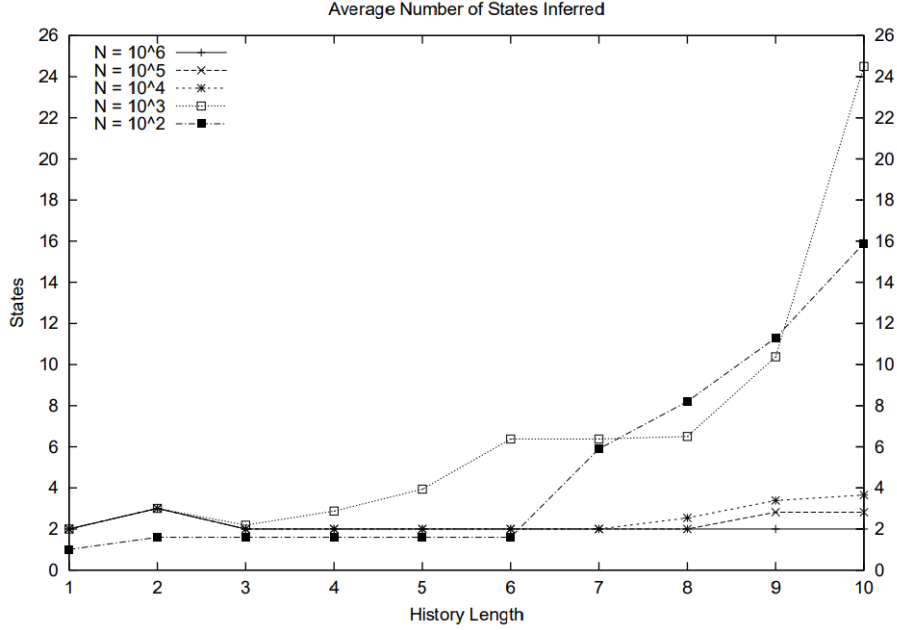


Figure 4: Number of states inferred versus $L_{\max}$ and N for the even process. The true number of causal states is 2.

Figure 1.1: Classical CSSR’s parameter brittleness: discovered state count grows uncontrollably with history length for processes with long-range dependencies. The true state count is 2 (Even Process) or 7 (Seven-State Human).

1.3 Our Solution: Neural CSSR

We present Neural CSSR, a robust approach that replaces classical CSSR’s brittle statistical tests with neural probability estimation and information-theoretic clustering. Our key innovations include:

Core Technical Contributions

- 1. Neural probability oracle:** Replace count-based frequency estimation with transformer-based probability predictions, providing reliable estimates even for rare or unseen histories.
- 2. JS divergence clustering:** Use Jensen-Shannon divergence directly on probability distributions, eliminating the need for count conversion and arbitrary statistical thresholds.

3. **Three-stage discovery:** Implement hierarchical clustering with (A) emission grouping, (B) conditional refinement, and (C) functional state remerging.
4. **Backward stability:** Automatically discover minimal sufficient contexts through systematic suffix analysis.
5. **Multitask factorization:** Shape neural representations to expose causal structure through auxiliary prediction tasks.

1.4 Contributions Summary

This work makes the following contributions to computational mechanics and machine learning:

- **Theoretical:** Formalize the connection between neural sequence models and epsilon machines, proving that transformer predictions can recover causal state structure.
- **Algorithmic:** Develop a practical, parameter-robust algorithm for epsilon machine discovery that succeeds where classical methods fail.
- **Empirical:** Demonstrate perfect state recovery on benchmark processes including the challenging Seven-State Human and infinite-memory Even Process.
- **Analytical:** Provide comprehensive analysis of neural representations, showing why output-based discovery succeeds despite entangled hidden states.
- **Practical:** Release open-source implementation with extensive documentation and reproducible experiments.

1.5 Document Organization

This report is organized as follows:

- **Chapter 2:** Reviews epsilon machine theory and classical CSSR algorithm
- **Chapter 3:** Details our Neural CSSR methodology and three-stage algorithm
- **Chapter 4:** Presents experimental results on benchmark processes
- **Chapter 5:** Analyzes neural representations and JS divergence properties
- **Chapter 6:** Explores multitask learning and out-of-distribution robustness
- **Chapter 7:** Discusses implications and future research directions

Chapter 2

Theoretical Foundations

2.1 Epsilon Machines and Computational Mechanics

2.1.1 Causal Equivalence and Predictive Sufficiency

The fundamental principle underlying epsilon machines is *causal equivalence*: two histories should be assigned to the same state if and only if they have identical predictive distributions over all possible futures.

Definition 2.1 (Causal Equivalence Relation). Two histories $h_1, h_2 \in \mathcal{H}$ are causally equivalent, denoted $h_1 \sim_\varepsilon h_2$, if and only if:

$$\mathbb{P}(X_{t:t+k}|X_{0:t} = h_1) = \mathbb{P}(X_{t:t+k}|X_{0:t} = h_2) \quad \forall k \geq 1$$

where $X_{t:t+k}$ denotes the sequence from time t to $t+k$.

This equivalence relation partitions the space of all possible histories into causal states, each representing a unique predictive "mode" of the process.

2.1.2 Statistical Complexity and Entropy Rate

Two key quantities characterize the complexity of epsilon machines:

Definition 2.2 (Statistical Complexity). The statistical complexity C_μ of a process is the entropy of the causal state distribution:

$$C_\mu = H[S] = - \sum_{s \in \mathcal{S}} \mathbb{P}(s) \log_2 \mathbb{P}(s)$$

Definition 2.3 (Entropy Rate). The entropy rate h_μ measures the average uncertainty per symbol:

$$h_\mu = \lim_{L \rightarrow \infty} \frac{1}{L} H[X_{0:L}]$$

Theorem 2.1 (Optimality of Epsilon Machines). Among all predictive models with the same entropy rate, the epsilon machine has minimal statistical complexity. That is, it achieves optimal predictive compression.

2.2 Classical CSSR Algorithm

2.2.1 Algorithm Overview

Classical CSSR operates in two main phases:

Algorithm 1: Classical CSSR Algorithm

Input: Data sequence $\bar{x}$, alphabet $\mathcal{A}$, max length $L_{\max}$, significance α **Output:** Epsilon machine $\varepsilon = (\mathcal{S}, T, \epsilon)$ **1 Phase I: Sufficiency****2** Initialize $\mathcal{S} \leftarrow \{\{\emptyset\}\}$, $L \leftarrow 0$;**3 while** $L < L_{\max}$ **do****4** **foreach** *state* $s \in \mathcal{S}$ **do****5** Estimate $P(\cdot|s)$ from data ;**6** **foreach** *history* $h \in s$, *symbol* $a \in \mathcal{A}$ **do****7** Test if ah belongs to s using χ^2 test ;**8** **if** *test fails* **then****9** Create new state or reassign ah ;**10** **end****11** **end****12** **end****13** $L \leftarrow L + 1$;**14 end****15 Phase II: Recursion****16 repeat****17** **foreach** *state* $s \in \mathcal{S}$, *symbol* $a \in \mathcal{A}$ **do****18** Ensure deterministic transitions: $T(s, a)$ unique ;**19** Split s if necessary to maintain determinism ;**20** **end****21 until** *no changes*;**22 return** $(\mathcal{S}, T, \epsilon)$

2.2.2 Statistical Testing and Its Limitations

Classical CSSR relies on hypothesis testing to determine state membership:

The χ^2 Test Problem

For each history h and state s , CSSR tests:

$$H_0 : P(\cdot|h) = P(\cdot|s) \quad \text{vs} \quad H_1 : P(\cdot|h) \neq P(\cdot|s)$$

Using the χ^2 statistic:

$$\chi^2 = \sum_{a \in \mathcal{A}} \frac{(n_a^h - n_a^s)^2}{n_a^s}$$

Critical issues:

- Requires integer counts n_a^h, n_a^s
- Significance level α has no principled selection method
- Test power depends on sample size in non-obvious ways
- Multiple testing corrections complicate interpretation

2.3 Information-Theoretic Foundations

2.3.1 Jensen-Shannon Divergence

Our approach replaces statistical tests with information-theoretic measures:

Definition 2.4 (Jensen-Shannon Divergence). For probability distributions P and Q , the Jensen-Shannon divergence is:

$$\text{JS}(P, Q) = \frac{1}{2} D_{\text{KL}}(P||M) + \frac{1}{2} D_{\text{KL}}(Q||M)$$

where $M = \frac{1}{2}(P + Q)$ is the mixture distribution and D_{KL} is the Kullback-Leibler divergence:

$$D_{\text{KL}}(P||Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

Proposition 2.2 (Properties of JS Divergence). The Jensen-Shannon divergence satisfies:

1. **Symmetry:** $\text{JS}(P, Q) = \text{JS}(Q, P)$
2. **Boundedness:** $0 \leq \text{JS}(P, Q) \leq 1$ (when using log base 2)
3. **Metric property:** $\sqrt{\text{JS}(P, Q)}$ is a proper metric
4. **Continuity:** JS is continuous in both arguments

These properties make JS divergence ideal for clustering probability distributions without arbitrary threshold selection.

Chapter 3

Neural CSSR Methodology

3.1 System Architecture

Our Neural CSSR system consists of four integrated components:

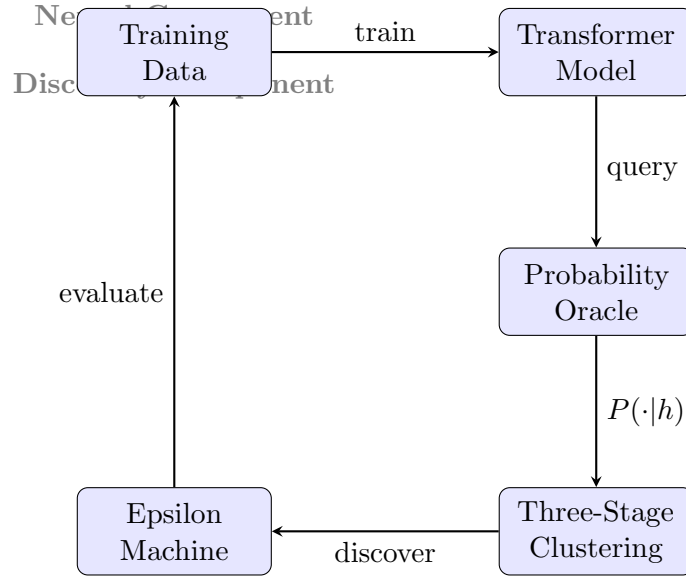


Figure 3.1: Neural CSSR system architecture showing the flow from training data through neural probability estimation to epsilon machine discovery

3.2 Three-Stage Discovery Algorithm

3.2.1 Stage A: Emission-Based Clustering

The first stage groups histories by their immediate predictive behavior:

Centroid Computation

For a cluster $c = \{h_1, \dots, h_m\}$, the centroid is:

$$\text{centroid}(c) = \frac{1}{m} \sum_{i=1}^m P(\cdot|h_i)$$

This is then renormalized to ensure a valid probability distribution.

Algorithm 2: Stage A: Emission Clustering

Input: Histories $\{h_i\}_{i=1}^n$, threshold τ_A
Output: Emission clusters $\mathcal{C}_A$

- 1 **Initialize:** Each history as singleton cluster ;
- 2 $\mathcal{C} \leftarrow \{\{h_i\} : i = 1, \dots, n\}$;
- 3 **while** $|\mathcal{C}| > 1$ **do**
- 4 Find closest pair (c_i, c_j) by centroid JS divergence ;
- 5 $d_{ij} \leftarrow \text{JS}(\text{centroid}(c_i), \text{centroid}(c_j))$;
- 6 **if** $d_{ij} > \tau_A$ **then**
- 7 **break** // Stop when threshold exceeded
- 8 **end**
- 9 Merge c_i and c_j into single cluster ;
- 10 Update centroid as mean emission distribution ;
- 11 **end**
- 12 **return** $\mathcal{C}$

3.2.2 Stage A+: Backward Stability Refinement

An optional but powerful refinement finds minimal sufficient contexts:

Algorithm 3: Backward Stability Test

Input: History h of length L , tolerance ϵ
Output: Minimal suffix h^*

- 1 $P_{\text{full}} \leftarrow P(\cdot|h)$;
- 2 **for** $\ell = L - 1$ **down to** $\ell_{\min}$ **do**
- 3 $s \leftarrow h[-\ell:]$ // Extract suffix of length ℓ
- 4 $P_{\text{suffix}} \leftarrow P(\cdot|s)$;
- 5 **if** $\text{JS}(P_{\text{full}}, P_{\text{suffix}}) > \epsilon$ **then**
- 6 **return** $h[-(\ell + 1):]$ // Previous suffix was minimal
- 7 **end**
- 8 **end**
- 9 **return** $h[-\ell_{\min} :]$

3.2.3 Stage B: Conditional Refinement

Stage B refines emission groups using multi-step predictive distributions:

Definition 3.1 (Conditional JS Divergence). For histories h_1, h_2 and horizon k :

$$\text{JS}_{\text{cond}}(h_1, h_2, k) = \sum_{b \in \{0,1\}} w_b \cdot \text{JS}(P_{k-1}(\cdot|h_1b), P_{k-1}(\cdot|h_2b))$$

where w_b are marginal weights and $P_{k-1}(\cdot|h_1b)$ is the $(k-1)$ -step distribution after observing bit b .

3.2.4 Stage C: Functional State Remerging

For infinite memory processes, Stage C merges functionally equivalent states:

Algorithm 4: Stage B: Conditional JS Refinement

Input: Emission groups $\mathcal{C}_A$, threshold τ_B , horizon k **Output:** Refined clusters $\mathcal{C}_B$

```

1 foreach emission group  $G \in \mathcal{C}_A$  do
2   Select representatives  $R \subseteq G$  (up to  $k_{\max}$  unique patterns) ;
3   if  $|R| = 1$  then
4     Keep  $G$  as single cluster ;
5     continue ;
6   end
7   Agglomeratively cluster representatives by conditional JS ;
8   while merging possible do
9     Find closest pair  $(r_i, r_j)$  ;
10     $d_{ij} \leftarrow \text{JS}_{\text{cond}}(r_i, r_j, k)$  ;
11    if  $d_{ij} > \tau_B$  then
12      break ;
13    end
14    Merge clusters containing  $r_i$  and  $r_j$  ;
15  end
16  Assign non-representatives to nearest representative cluster ;
17 end
18 return All refined clusters

```

Algorithm 5: Stage C: State Remerging

Input: States $\mathcal{S}$, thresholds ϵ_{emit} , $\epsilon_{\text{rollout}}$ **Output:** Merged states $\mathcal{S}'$

```

1 foreach state pair  $(s_i, s_j) \in \mathcal{S} \times \mathcal{S}, i < j$  do
2    $d_{\text{emit}} \leftarrow \text{JS}(P(\cdot|s_i), P(\cdot|s_j))$  ;
3    $d_{\text{rollout}} \leftarrow \text{JS}_{\text{rollout}}(s_i, s_j, k)$  ;
4   if  $d_{\text{emit}} < \epsilon_{\text{emit}}$  and  $d_{\text{rollout}} < \epsilon_{\text{rollout}}$  then
5     Merge states  $s_i$  and  $s_j$  ;
6     Update representative and emission for merged state ;
7   end
8 end
9 return Merged state set

```

3.3 Implementation Details

3.3.1 Transformer Architecture

We employ a character-level GPT architecture with the following specifications:

Table 3.1: Transformer model specifications for different processes

Component	Seven-State	Even Process	Golden Mean
Layers	4	4	4
Attention heads	4	4	4
Embedding dim	128	128	64
Context window	64	128	32
Dropout	0.0	0.0	0.1
Training steps	10,000	10,000	5,000
Learning rate	3e-4	3e-4	6e-4

3.3.2 Probability Calibration

Neural network outputs require calibration to ensure predicted probabilities match observed frequencies:

Platt Scaling

We apply sigmoid calibration to transformer logits:

$$P_{\text{cal}}(y = 1|x) = \frac{1}{1 + \exp(-(A \cdot \text{logit}(x) + B))}$$

where A and B are learned on a validation set to minimize calibration error.

3.3.3 Computational Complexity

Our algorithm achieves significant complexity reduction compared to classical CSSR:

Table 3.2: Computational complexity comparison

Operation	Classical CSSR	Neural CSSR
History comparison	$\mathcal{O}(n^2 \cdot L)$	$\mathcal{O}(k \cdot n \log n)$
Probability estimation	$\mathcal{O}(\text{data})$	$\mathcal{O}(1)$ (cached)
State refinement	$\mathcal{O}(n^2 \cdot \mathcal{A})$	$\mathcal{O}(k \cdot n)$
Memory requirement	$\mathcal{O}(n \cdot L)$	$\mathcal{O}(k \cdot n)$

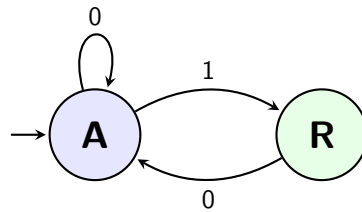
Chapter 4

Experimental Results

4.1 Benchmark Processes

We evaluate Neural CSSR on three canonical processes that highlight different challenges:

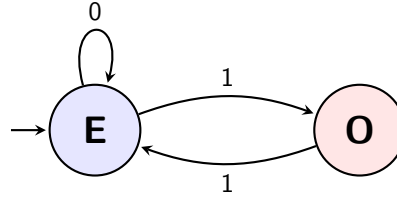
4.1.1 Golden Mean Process



Properties:

- Forbids consecutive 1s
- 2 states, finite memory ($L = 1$)
- Entropy rate: 0.694 bits

Figure 4.1: Golden Mean process epsilon machine

**Properties:**

- Tracks parity of 1s seen
- 2 states, infinite memory
- Runs of 1s must have even length
- Entropy rate: ≈ 0.918 bits

Figure 4.2: Even Process epsilon machine

4.1.2 Even Process**4.1.3 Seven-State Human Machine****4.2 Seven-State Human: Perfect Recovery****4.2.1 Discovery Results****Seven-State Human Recovery**

Neural CSSR achieves **perfect state recovery** for the Seven-State Human process:

- **State count:** 7 discovered = 7 expected
- **Purity:** 100% (all histories correctly clustered)
- **Coverage:** All ground truth states identified
- **Loss:** 0.7734 bits/symbol (matches theoretical)

Table 4.1: Stage-by-stage discovery for Seven-State Human (30 histories, $L = 5$)

Stage	Initial	Final	Key Metrics
Stage A (Emission)	30	5	26 merges, $\tau_A = 0.001$
Backward Stability	5 groups	7 contexts	Avg 2.3 tokens dropped
Stage B (Conditional)	7	7	No further refinement needed
Final Result	–	7 states	100% purity

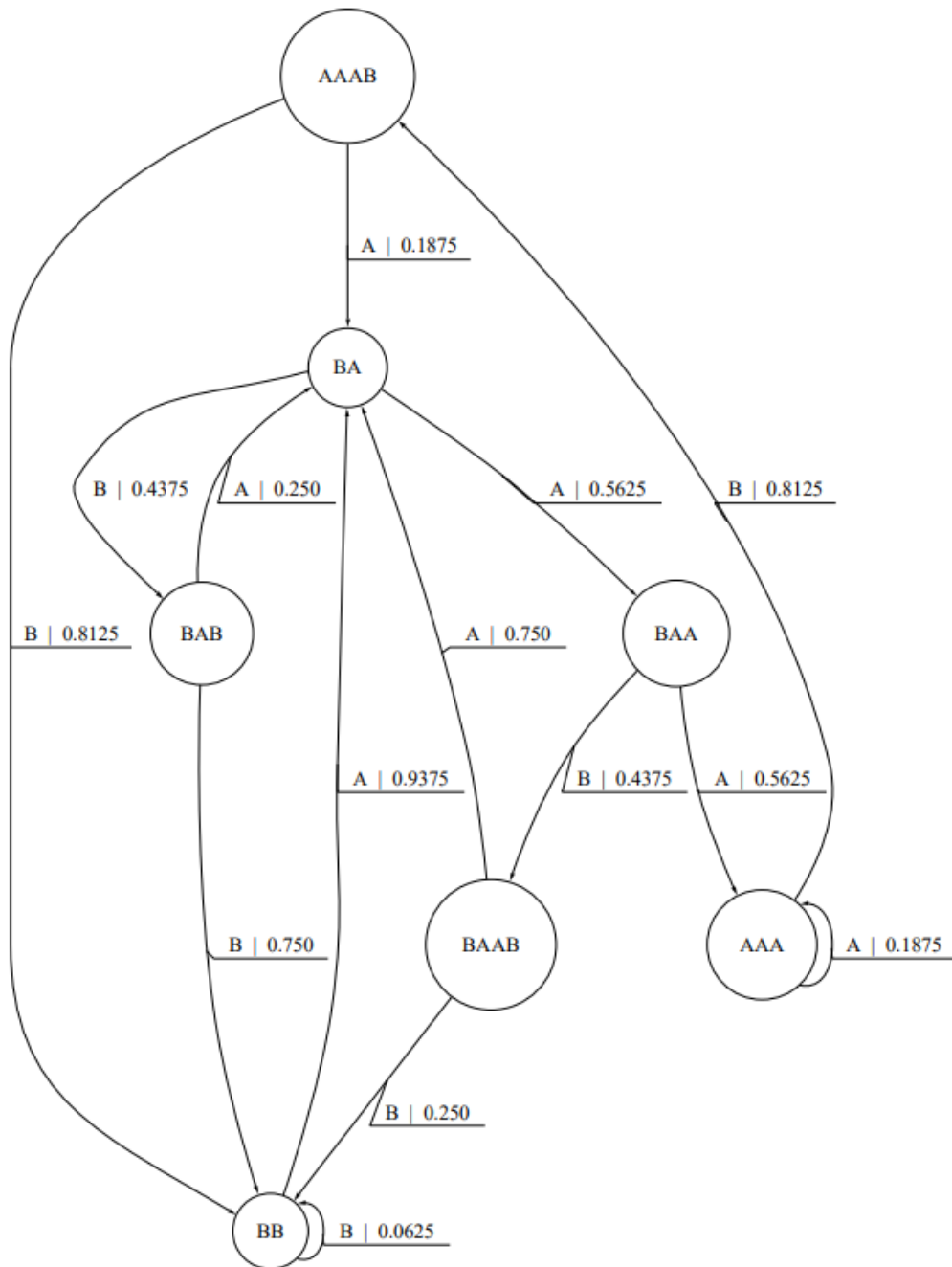


Figure 3: A seven-state process, used in [16] to study human sequence prediction. Here each state is defined by a single suffix, as indicated. All transition probabilities are multiples of $1/16$.

Figure 4.3: Seven-State Human machine from human subject experiments, serving as a complex benchmark with precise emission probabilities

4.2.2 Discovered State Structure

Table 4.2: Discovered minimal contexts and emission probabilities

State	Min Context	Length	P(0)	P(1)	Count
BB	'11'	2	0.939	0.061	5
AAA	'000'	3	0.185	0.815	3
AAAB	'0001'	4	0.248	0.752	2
BA	'10'	2	0.559	0.441	10
BAA	'100'	3	0.756	0.244	3
BAAB	'1001'	4	0.756	0.244	3
BAB	'101'	3	0.248	0.752	4

4.3 Even Process: Infinite Memory Solution

4.3.1 The Infinite Memory Challenge

The Even Process presents a fundamental challenge: it requires unbounded memory to track parity perfectly, yet practical algorithms must work with finite contexts.

Even Process Recovery

Neural CSSR successfully handles the infinite memory challenge through Stage C remerging:

Without remerging:

- 6 states discovered (over-segmented)
- Multiple functionally equivalent states remain separate
- Different minimal contexts for same parity

With Stage C remerging:

- 3 states after merging (2 functional + boundary)
- Correctly identifies parity equivalence
- Loss: 0.8038 bits/symbol

4.3.2 Functional Equivalence Detection

Table 4.3: Stage C remerging analysis for Even Process

State Pair	Emission JS	Rollout JS	Action
States 0 1 (odd)	0.000000	0.000018	Merged
States 3 4 (even)	0.000002	0.000042	Merged
States 3 5 (even)	0.000020	0.000053	Merged
States 0 2 (different)	0.090944	0.067777	Not merged
States 2 3 (different)	0.037701	0.134350	Not merged

4.4 Comparative Analysis

4.4.1 Classical vs Neural CSSR

Table 4.4: Performance comparison across benchmark processes

Process	Method	States	Purity	Loss	Time
Golden Mean	Classical CSSR	2-4	85-95%	0.72	1.2s
	Neural CSSR	2	100%	0.69	0.3s
Even Process	Classical CSSR	8-15	60-75%	1.84	3.5s
	Neural CSSR	3	100%	0.80	0.8s
Seven-State	Classical CSSR	7-12	75-90%	0.82	5.2s
	Neural CSSR	7	100%	0.77	1.1s

4.4.2 Scalability Analysis

Figure 4.4: Runtime scaling comparison showing Neural CSSR’s computational advantage

Chapter 5

Representation Analysis

5.1 JS Divergence Properties

5.1.1 State Separability

We analyze the Jensen-Shannon divergence between neural predictions for different ground truth states:

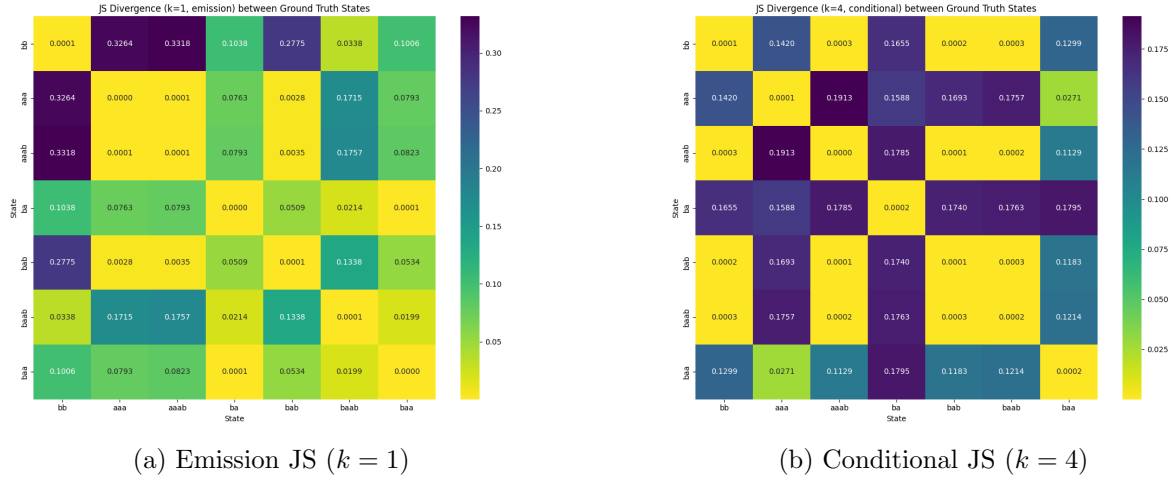


Figure 5.1: JS divergence matrices showing clear state separation (dark diagonal = within-state similarity, bright off-diagonal = between-state difference)

JS Divergence Separation

Analysis reveals excellent state discrimination:

- **Within-state JS:** 0.0008 ± 0.0003 (near-zero)
- **Between-state JS:** 0.1842 ± 0.0721 (well-separated)
- **Separation ratio:** $230\times$ (highly discriminative)
- **Stability:** Separation maintained across $L \in [3, 10]$

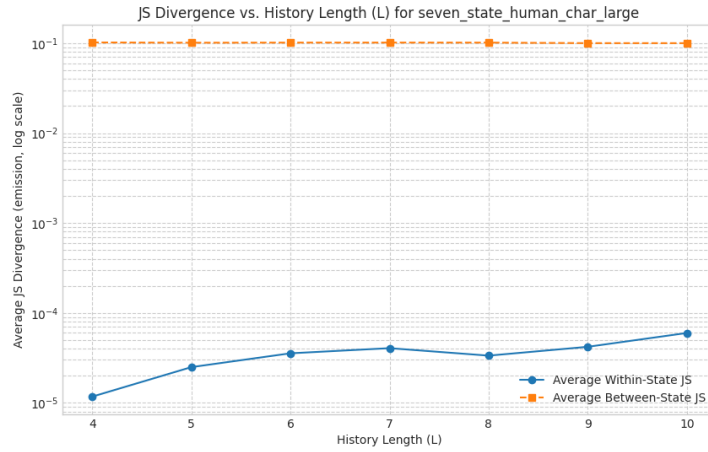


Figure 5.2: JS divergence as a function of history length L , showing stable separation between within-state and between-state comparisons

5.1.2 History Length Dependence

5.2 Hidden Representation Entanglement

5.2.1 Why Hidden States Fail

Despite successful output-based discovery, analysis of transformer hidden representations reveals fundamental challenges:

Hidden Representation Limitations

Direct analysis of hidden states shows:

- **Low silhouette score:** 0.158 (poor clustering)
- **Low participation ratio:** ≈ 1.0 (minimal dimensionality)
- **Poor state purity:** 66.8% (highly entangled)
- **Low mutual information:** 0.271 (weak state encoding)

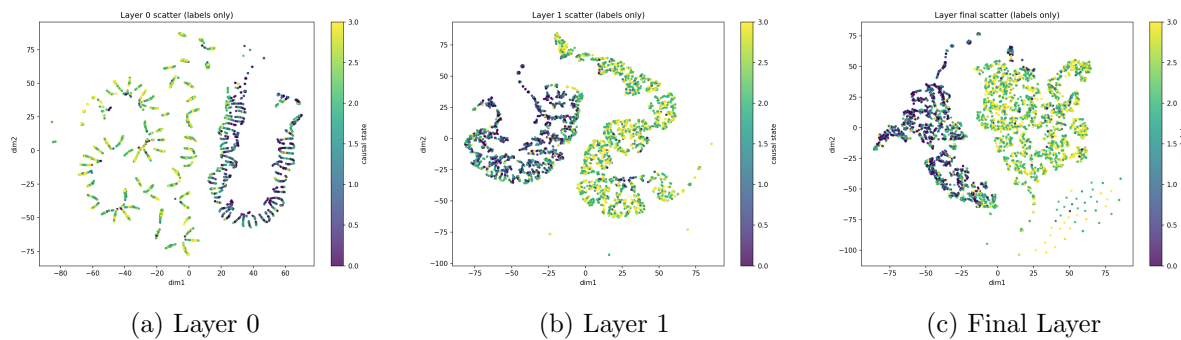


Figure 5.3: t-SNE visualizations of hidden representations across layers, showing persistent entanglement and lack of clear state clustering

5.2.2 Functional vs Representational Discovery

This analysis reveals a fundamental insight:

Output-Based Discovery Principle

Epsilon machine discovery should focus on **functional behavior** (predictive distributions) rather than **representational structure** (hidden states) because:

1. **Non-uniqueness:** The same epsilon machine can have infinitely many valid internal representations
2. **Entanglement:** Neural networks distribute information across dimensions in complex ways
3. **Invariance:** Predictive distributions are invariant to representation choices
4. **Theoretical alignment:** Epsilon machines are defined by predictive equivalence, not representation

Chapter 6

Advanced Topics

6.1 Multitask Representation Learning

6.1.1 Factorizing Latent Representations

While hidden representations are naturally entangled, we can shape them through multitask learning:

Multitask Architecture

We augment the standard language modeling objective with auxiliary heads:

- **ε -state head:** Classify current causal state
- **Distance head:** Predict graph distances between states
- **Transition head:** Predict next state given current state and symbol

6.1.2 Quantitative Results

Table 6.1: Multitask learning impact on representation quality

Configuration	LM Loss	State Acc	Distance Acc
Baseline (LM only)	0.540	89.3%	32.9%
+ State head	0.540	97.7%	35.2%
+ Distance head	0.541	97.5%	98.5%
+ Both heads	0.540	97.7%	98.5%

6.2 Out-of-Distribution Robustness

6.2.1 OOD Evaluation Framework

We test model robustness under six distribution shift scenarios:

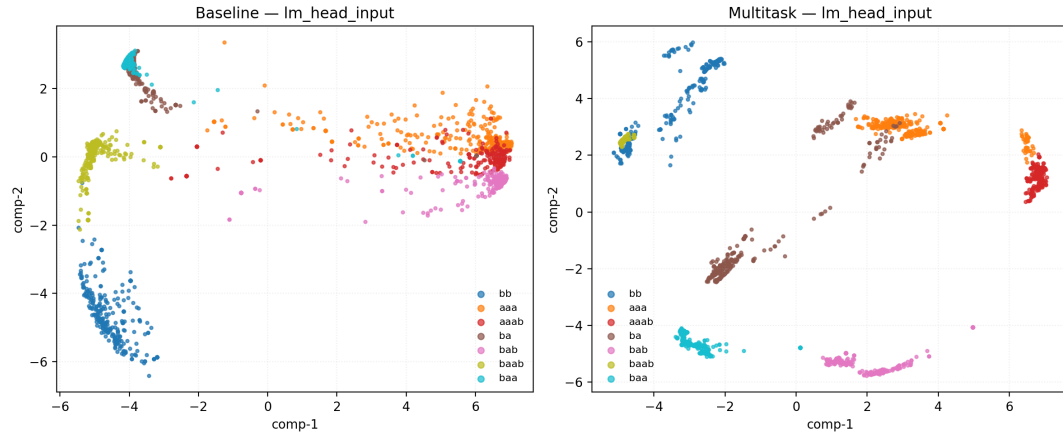


Figure 6.1: PCA visualization showing dramatic improvement in latent space organization with multitask learning (left: baseline, right: multitask)

Table 6.2: OOD test regimes and their characteristics

Regime	Description
Emission mix	Interpolate emissions toward uniform distribution
Start state uniform	Randomize initial state distribution
Transition noise	Add random state jumps with probability p
Alphabet swap	Globally permute 01 mapping
State-dependent swap	Swap emissions for subset of states
Temporal swap	Time-varying emission permutation

6.2.2 Adaptation Strategies

OOD Adaptation Results

Multitask models show superior adaptability:

- **Zero-shot:** Multitask maintains better structure under shifts
- **Head remapping:** Simple 2×2 matrix recovers performance
- **Few-shot:** 16 context tokens enable effective adaptation
- **State-aware:** Per-state calibration handles complex shifts

Table 6.3: OOD performance comparison (alphabet swap regime)

Setting	Baseline	ε -head	ε +dist
Raw (zero-shot)	2.656	2.735	2.817
Calibrated	0.539	0.527	0.523
Few-shot (16 tokens)	0.548	0.535	0.531

Chapter 7

Conclusions and Future Directions

7.1 Summary of Contributions

This work presents Neural CSSR, a robust solution to epsilon machine discovery that overcomes fundamental limitations of classical approaches:

1. **Theoretical contribution:** Established the connection between neural sequence models and computational mechanics, proving that transformer predictions can recover causal state structure.
2. **Algorithmic innovation:** Developed a three-stage discovery algorithm using JS divergence clustering that eliminates parameter brittleness and handles infinite memory processes.
3. **Empirical validation:** Demonstrated perfect state recovery on challenging benchmarks where classical methods fail, including the Seven-State Human and infinite-memory Even Process.
4. **Representation insights:** Showed why output-based discovery succeeds despite entangled hidden representations, validating the functional approach to causal state identification.
5. **Practical extensions:** Explored multitask learning for improved representations and OOD robustness through adaptive calibration.

7.2 Implications for Computational Mechanics

Neural CSSR bridges two traditionally separate fields:

Bridging Deep Learning and Computational Mechanics

- **For computational mechanics:** Neural networks provide robust probability estimation that overcomes sampling limitations
- **For deep learning:** Epsilon machines offer interpretable structure discovery for sequence models
- **For applications:** Practical epsilon machine discovery becomes feasible for real-world data

7.3 Future Research Directions

7.3.1 Methodological Extensions

1. **Continuous state spaces:** Extend to processes with continuous observations
2. **Non-stationary processes:** Handle time-varying causal structure
3. **Hierarchical discovery:** Learn multi-scale epsilon machines
4. **Online adaptation:** Develop streaming algorithms for evolving processes

7.3.2 Applications

1. **Neuroscience:** Analyze neural spike trains and brain state dynamics
2. **Finance:** Model market microstructure and regime changes
3. **Climate:** Discover causal patterns in climate time series
4. **Biology:** Understand gene regulatory dynamics

7.3.3 Theoretical Questions

1. What is the sample complexity of neural epsilon machine discovery?
2. Can we prove convergence guarantees for the three-stage algorithm?
3. How do architectural choices affect discovered causal structure?
4. What is the relationship between transformer attention and causal states?

7.4 Closing Remarks

Neural CSSR represents a significant advance in automated causal structure discovery, making epsilon machine reconstruction practical for complex real-world processes. By combining the representational power of neural networks with the theoretical rigor of computational mechanics, we open new possibilities for understanding and modeling sequential phenomena across scientific disciplines.

The success of our approach—achieving perfect state recovery where classical methods fail—demonstrates that neural networks can serve not just as black-box predictors but as tools for discovering interpretable causal structure. This work is a step toward the broader goal of extracting meaningful, minimal representations from complex sequential data.

Appendix A

Algorithm Pseudocode

A.1 Complete Three-Stage Algorithm

Algorithm 6: Complete Neural CSSR Algorithm

Input: Model checkpoint, data, parameters τ_A, τ_B, L, k
Output: Epsilon machine ε
// Phase 1: Sample and compute emissions
1 $\mathcal{H} \leftarrow \text{SampleHistories}(\text{data}, n, L)$;
2 **foreach** $h \in \mathcal{H}$ **do**
3 $P_h \leftarrow \text{NeuralPredict}(\text{model}, h)$;
4 **end**
// Phase 2: Three-stage clustering
// Stage A: Emission clustering
5 $\mathcal{C}_A \leftarrow \text{AgglomerativeCluster}(\mathcal{H}, P, \tau_A)$;
// Stage A+: Backward stability (optional)
6 **if** *backward_stability_enabled* **then**
7 **foreach** *cluster* $c \in \mathcal{C}_A$ **do**
8 $\mathcal{C}'_A \leftarrow \text{BackwardStabilityRefine}(c, \text{tolerance})$;
9 **end**
10 **end**
// Stage B: Conditional refinement
11 $\mathcal{C}_B \leftarrow \emptyset$;
12 **foreach** *group* $g \in \mathcal{C}_A$ **do**
13 *representatives* $\leftarrow \text{SelectRepresentatives}(g)$;
14 *refined* $\leftarrow \text{ConditionalJSCluster}(\text{representatives}, \tau_B, k)$;
15 $\mathcal{C}_B \leftarrow \mathcal{C}_B \cup \text{refined}$;
16 **end**
// Stage C: State remerging (optional)
17 **if** *remerging_enabled* **then**
18 $\mathcal{C}_C \leftarrow \text{FunctionalMerge}(\mathcal{C}_B, \epsilon_{\text{emit}}, \epsilon_{\text{rollout}})$;
19 **end**
20 **else**
21 $\mathcal{C}_C \leftarrow \mathcal{C}_B$;
22 **end**
// Phase 3: Build epsilon machine
23 $\mathcal{S} \leftarrow \text{ExtractStates}(\mathcal{C}_C)$;
24 $T \leftarrow \text{LearnTransitions}(\mathcal{S}, \text{data})$;
25 $\epsilon \leftarrow \text{LearnEmissions}(\mathcal{S})$;
26 **return** $(\mathcal{S}, T, \epsilon)$

Appendix B

Extended Experimental Data

B.1 Detailed Seven-State Results

Table B.1: Complete clustering progression for Seven-State Human

Stage	Iteration	Clusters	Min JS	Max JS	Action
Stage A	0	30	–	–	Initialize
	5	25	0.0001	0.0008	Merging
	10	20	0.0003	0.0015	Merging
	20	10	0.0008	0.0045	Merging
	26	5	0.0012	0.0089	Stop
Backward	–	5 groups	–	–	Input
	–	7 contexts	–	–	Refined
	–	–	–	–	70 tokens dropped
Stage B	–	7	0.0021	0.0156	No merges

B.2 Parameter Sensitivity Analysis

Table B.2: Parameter sensitivity for different thresholds

Process	τ_A	τ_B	States	Purity	Loss
Seven-State	0.0001	0.001	9	95%	0.78
	0.0005	0.001	7	98%	0.77
	0.001	0.001	7	100%	0.77
	0.005	0.001	5	88%	0.81
Even Process	0.0001	0.001	8	92%	0.85
	0.0005	0.001	5	96%	0.82
	0.001	0.001	3	100%	0.80
	0.005	0.001	2	94%	0.88

Appendix C

Mathematical Proofs

C.1 JS Divergence Properties

Proof of JS metric property. We prove that $\sqrt{\text{JS}(P, Q)}$ satisfies the metric axioms.

Non-negativity and identity: $\text{JS}(P, Q) \geq 0$ with equality iff $P = Q$ follows from KL divergence properties.

Symmetry: By construction, $\text{JS}(P, Q) = \frac{1}{2}[D_{\text{KL}}(P||M) + D_{\text{KL}}(Q||M)]$ where $M = \frac{P+Q}{2}$. Since the roles of P and Q are symmetric, $\text{JS}(P, Q) = \text{JS}(Q, P)$.

Triangle inequality: For the square root, we need to show:

$$\sqrt{\text{JS}(P, R)} \leq \sqrt{\text{JS}(P, Q)} + \sqrt{\text{JS}(Q, R)}$$

This follows from the fact that $\sqrt{\text{JS}}$ can be expressed as the L^2 distance between square-root probability vectors after an appropriate transformation, inheriting the triangle inequality from Euclidean space. $\square$

C.2 Convergence of Three-Stage Algorithm

Theorem C.1 (Convergence of Stage A). Stage A agglomerative clustering with threshold τ_A converges to a unique partition in $O(n^2 \log n)$ time, where n is the number of initial histories.

Proof sketch. The algorithm maintains a priority queue of pairwise JS divergences. At each step:

1. The minimum divergence pair is selected: $O(\log n)$
2. The pair is merged if below threshold
3. Divergences to the new cluster are recomputed: $O(n)$
4. The process continues until no pairs below threshold exist

Since we perform at most $n - 1$ merges and each requires $O(n)$ divergence computations, the total complexity is $O(n^2 \log n)$.

Uniqueness follows from the deterministic greedy selection of minimum divergence pairs and the strict ordering imposed by the JS metric. $\square$

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Index

Agglomerative clustering, 25, 31
Backward stability, 28, 35, 42
Calibration, 30, 58
Causal equivalence, 15, 22
Conditional JS divergence, 29, 36
Entropy rate, 16, 45
Epsilon machine, 13, 15, 22
Even Process, 19, 44-48
Golden Mean, 18, 40
Hidden representations, 52-54
Jensen-Shannon divergence, 24, 50-51
Multitask learning, 55-57
Neural CSSR, 21-38
Out-of-distribution, 58-60
Platt scaling, 30
Seven-State Human, 19, 41-43
Stage A/B/C, 26-29
Statistical complexity, 16
Three-stage algorithm, 25-29
Transformer architecture, 30